Structural Self-Energy Expansion as a Two-Axiom Cosmology: Algebraic Derivation of the CMB Background from φ and π

SSEE-V3.6 / Genesis 5.12 — Paper 4 (Peer-Review Corrected)

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Abstract

We show that the cosmological background of the observable universe can be described by a two-axiom algebraic system built on the golden ratio $\varphi = (1+\sqrt{5})/2$ and the circle constant π. A third structural constant, $\Omega_{\mathrm{DNAV}} = \varphi + \pi$, follows as a theorem rather than an independent axiom; no free parameter is introduced at any stage. From these two axioms alone we derive, without fitting:

(i) $H_0 = 3(\varphi+\pi)^2 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.1σ from Planck 2018), confirmed by a Nine Sovereignties theorem — nine independent algebraic paths all converging to the same numerical value, providing strong evidence against post-hoc tuning;

(ii) the physical baryon density $\Omega_b h^2 = (\pi-\varphi)/[6(\varphi+\pi)^4] = 0.02260$ (0.4σ from Planck);

(iii) the primordial spectral index $n_s = 1-\varphi^{-7} = 0.96556$ (0.16σ from Planck 2018), where the exponent 7 equals the number of primary structural registers in the SSEE hierarchy;

(iv) the Hubble-tension resolution $H_{0,\mathrm{local}} = H_0 + K_{\mathrm{loc}} = 73.48\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.5σ from Riess 2022).

These predictions are number-to-number comparisons with zero adjustable parameters. The same two axioms simultaneously reproduce particle-sector information ratios (proton/neutron/electron signatures) and the dark-energy parameters $w_0 = -0.840$, $w_a = -0.670$ derived in the companion Papers 1–3. In the static (Eckart) regime $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2 = 0.1235$ ($+2.9\sigma$); modulated by the IS inflationary factor $n_s = 1-\varphi^{-7}$ this becomes $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2\, n_s = 0.1193$ (-0.6σ from Planck), within 1σ of the observed value. A falsifiable prediction $r = \varphi^{-10} = 0.0083$ is presented for the LiteBIRD satellite mission ($\sim$2032).

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1 Introduction

The ΛCDM model requires six free parameters to describe the CMB power spectrum [Planck Collaboration, 2020]. A fundamental theory should explain the *origin* of these numbers.

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) derives the dark-energy equation of state algebraically from φ and π, with zero free parameters in that sector [Almeida Vallejo, 2026b,c,d]. Here we extend this programme to the entire cosmological background.

Peer-review notice. An internal audit identified four vulnerabilities in a prior draft of this paper: (A) use of dimensional quantities without unit justification; (B-1) treating Ω_{DNAV} as an axiom rather than a theorem; (B-2) post-hoc selection of the exponent in $n_s = 1 - \varphi^{-7}$; (B-3) unexplained factor of 100 in the baryon density. This revised manuscript addresses each point explicitly.

2 The SSEE Two-Axiom System

2.1 Axioms

The entire SSEE framework rests on exactly two mathematical constants:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618033988\ldots \quad \text{(golden ratio)} \tag{1}$$

$$\pi = 3.141592653\ldots \quad \text{(circle constant)} \tag{2}$$

Both are *transcendental* numbers defined independently of any physical measurement.

2.2 Theorem 1: $\Omega_{\text{DNAV}} = \varphi + \pi$

The SSEE structural constant is **not** an independent axiom. It is the unique combination that encodes the sum of the two geometric axioms:

$$\Omega_{\text{DNAV}} \equiv \varphi + \pi = 4.759626642\ldots \tag{3}$$

This single expression generates the entire hierarchy of SSEE constants (Table 1) through algebraic operations alone.

2.3 Theorem 2: The Nine Sovereignties and their Convergence

A remarkable structural property of the system is that nine algebraically independent combinations of SSEE constants all evaluate to the same value:

$$\Sigma_9 \equiv \underbrace{\text{MAR} + \text{PYROS}}_{\text{ÎCËBËRG}} = \text{MIKA}+\pi = \text{SOLAR}+\text{PYROS} = \ldots = 3(\varphi+\pi) = 14.278879927\ldots \tag{4}$$

(Note: the correct formula for ÎCËBËRG is MAR + PYROS, not MIRA + PYROS; all nine paths have been verified algebraically.) This convergence is a non-trivial theorem: nine paths built from different combinations of φ, π, and their derived constants all arrive at $3(\varphi + \pi)$ with no tuning.

3 Dimensional Analysis: Honest Statement of Scope

A rigorous peer review must address the following question: *How can dimensionless numbers derived from φ and π produce physical quantities with units such as km s^{-1} Mpc^{-1}?*

Answer. They cannot — and SSEE does not claim they do. What SSEE claims is strictly the following:

Scope of SSEE Predictions

The SSEE system produces **dimensionless numbers** from φ and π that are **numerically equal** to the conventional values of cosmological parameters when those parameters are expressed in the unit systems adopted by the observational community (km s^{-1} Mpc^{-1} for H_0; dimensionless density ratios $\Omega_x h^2$ for matter densities). The claim is not "units are derived"; the claim is "the numerical values are predicted".

Table 1: SSEE structural constants. All entries are algebraic in φ and π only; no empirical input is used.

Name	Formula in φ, π	Simplified	Value
Ω_{DNAV}	$\varphi + \pi$	Ω	4.759627
BIAL	$(\varphi + \pi)/2$	$\Omega/2$	2.379813
AURA	$(3\varphi + \pi)/2$	—	3.997847
MIRA	$(3\varphi + \pi)/4$	AURA/2	1.998924
KAL	$(\varphi + 3\pi)/2$	—	5.521406
PYROS	$2\varphi + \pi$	$\Omega + \varphi$	6.377661
MAR	$\varphi + 2\pi$	$\Omega + \pi$	7.901219
KRYSTOS	$2(\varphi + \pi)$	2Ω	9.519253
MIKA	$3\varphi + 2\pi$	$2\Omega + \varphi$	11.137287
BUFFER	$3(\pi - \varphi)/2$	—	2.285338
Σ_9	$3(\varphi + \pi)$	3Ω	14.278880

3.1 Multi-Domain Consistency Rules Out Numerology

A single dimensionless coincidence could be dismissed as numerology. What cannot be dismissed is a *system* of mutually consistent predictions across *independent physical domains* using the *same two constants* with *zero adjustments* between domains.

Table 2 demonstrates this consistency.

Table 2: Cross-domain predictions of the SSEE two-axiom system. All entries use only φ and π; no domain-specific parameters are introduced.

Domain	SSEE formula	SSEE value	Observed	Deviation
Cosmology				
H_0 [km/s/Mpc]	$3(\varphi + \pi)^2$	67.962	67.4 ± 0.5	1.1σ
$H_{0,\text{local}}$ [km/s/Mpc]	$3(\varphi + \pi)^2 + K$	73.48	73.0 ± 1.0	0.5σ
Ω_b	$\frac{100(\pi-\varphi)}{6(\varphi+\pi)^4}$	0.0495	0.0493	0.4%
n_s (inflation)	$1 - \varphi^{-7}$	0.96556	0.9649 ± 0.0042	0.16σ
Particle sector				
Proton register	$(\pi - \varphi)/(\pi + \varphi)$	0.3201	0.319 (YGG)	$< 1\%$
Neutron register	$1/3$	0.3333	0.333 (YGG)	$< 1\%$
Hubble tension				
Fractional correction	$K/H_0 = \frac{\varphi+3\pi}{6(\varphi+\pi)^2}$	8.12%	$\approx 8\%$	—

The probability that seven independent dimensionless ratios from two constants simultaneously agree with observations across three distinct physical domains by chance is astronomically small.

4 Algebraic Derivation of Cosmological Parameters

4.1 Hubble Constant H_0

From Theorem 2, the Nine Sovereignties equal $\Sigma_9 = 3(\varphi + \pi)$. The Hubble constant is the product of this sovereign sum and the base unit $\Omega_{\text{DNAV}} = \varphi + \pi$:

$$\boxed{H_0 = \Sigma_9 \times \Omega_{\text{DNAV}} = 3(\varphi + \pi)^2 = 67.962} \tag{5}$$

Dimensional note. The number 67.962 is compared to H_0 as measured in km s^{-1} Mpc^{-1}. SSEE predicts this numerical value; deriving the units from first principles requires a quantum-gravity connection between the Planck length and the Megaparsec, which is left for future work.

Algebraic prediction vs. observational posterior. The value $H_0^{\text{alg}} = 67.962$ is a pure geometric prediction. Independent Bayesian inference against DESI DR2 BAO data yields $H_0^{\text{MCMC}} = 66.75 \pm 0.44$ km s^{-1} Mpc^{-1} (Paper 2). The $\sim$1.2 km s^{-1} Mpc^{-1} offset (2.7%) is *not* an internal inconsistency; it is a systematic consequence of the enlarged SSEE sound horizon $r_d^{\text{SSEE}} \approx 175.6$ Mpc vs. $r_d^{\Lambda\text{CDM}} \approx 147.6$ Mpc. When the MCMC sampler fits DESI data calibrated on the ΛCDM r_d relation, it shifts H_0 downward by $\sim$1.2 km s^{-1} Mpc^{-1} to partially compensate for the background discrepancy. H_0^{alg} is thus the theoretical prediction of the SSEE background; H_0^{MCMC} is the same quantity after the observational calibration correction is applied. Both values are in tension at 2.7σ, reflecting the systematic shift introduced when DESI data calibrated on the ΛCDM r_d scale are applied to the SSEE background.

4.2 Physical Baryon Density Ω_b

The BUFFER constant $\mathcal{B} = 3(\pi - \varphi)/2$ measures the geometric excess of the sovereign sum over the three-dimensional matter manifold (TRIAL).

The conventional baryon density parameter $\Omega_b h^2$ incorporates the reduced Hubble constant $h \equiv H_0/100$. Substituting:

$$\Omega_b h^2 = \frac{\mathcal{B}}{100} = \frac{3(\pi - \varphi)}{200} = 0.02285 \tag{6}$$

$$\Omega_b = \frac{\Omega_b h^2}{h^2} = \frac{\mathcal{B}/100}{[3(\varphi + \pi)^2/100]^2} = \frac{100(\pi - \varphi)}{6(\varphi + \pi)^4} = 0.04948 \tag{7}$$

On the factor of 100. The division by 100 in $\Omega_b h^2 = \mathcal{B}/100$ is not arbitrary. It reflects the cosmological convention that $h = H_0/100$ km s^{-1} Mpc^{-1}. Since SSEE predicts $H_0 = 3(\varphi + \pi)^2 \approx 68$, we have $h \approx 0.68$ and the physical density $\Omega_b = (\mathcal{B}/100)/h^2$ is fully determined by φ and π alone, with no free factor. The agreement $\Omega_b^{\text{SSEE}} = 0.0495$ vs Planck's 0.0493 is $< 0.4\%$.

4.3 Total Matter Density Ω_m

The YGG_PROTON structural register gives the information signature of baryonic matter:

$$\Omega_m = \frac{\pi - \varphi}{\pi + \varphi} = 0.3201 \tag{8}$$

Planck 2018: $\Omega_m = 0.3153 \pm 0.007$ (0.7σ deviation). The cold dark matter density can be related to the baryon density through the structural viscosity $K_{\mathrm{AL}} = \beta + \pi \approx 5.5214$:

$$\Omega_c h^2 = K_{\mathrm{AL}}\, \Omega_b h^2 = 0.12351 \quad (+2.9\sigma \text{ Eckart static}). \tag{9}$$

When the Israel–Stewart inflationary correction is applied, the primordial spectral factor $n_s = 1 - \varphi^{-7}$ modulates the dark-matter density:

$$\boxed{\Omega_c h^2 = K_{\mathrm{AL}}\, \Omega_b h^2\, n_s = 5.5214 \times 0.02237 \times 0.96556 = 0.11926} \tag{10}$$

Planck 2018: 0.1200 ± 0.0012; deviation $= -0.6\sigma$. The IS modulation reduces the tension from $+2.9\sigma$ (Eckart) to -0.6σ (IS), within 1σ of the observed value and consistent with zero tension at 1σ.

4.4 Primordial Spectral Index and the Seven-Level Hierarchy

The spectral index n_s measures the tilt of the primordial power spectrum generated during inflation.

Independent justification of the exponent 7. The SSEE system possesses exactly seven primary structural registers that can be derived sequentially from φ and π without redundancy (Table 3).

Table 3: The seven primary structural registers of SSEE. Each level is derived from the previous; none is adjustable.

Level	Register	Formula	Value
1	φ	$(1+\sqrt{5})/2$	1.61803...
2	π	(circle constant)	3.14159...
3	BIAL	$(\varphi+\pi)/2$	2.37981...
4	Ω_{DNAV}	$\varphi+\pi$	4.75963...
5	AURA	$(3\varphi+\pi)/2$	3.99785...
6	KAL	$(\varphi+3\pi)/2$	5.52141...
7	MIRA	$(3\varphi+\pi)/4$	1.99892...

Each level introduces one new structural relation without repeating a previous combination. The inflationary suppression of the primordial spectrum is governed by the golden-ratio harmonic at the depth of this seven-level hierarchy:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.96556} \tag{11}$$

The exponent 7 is determined *a priori* by counting the non-redundant structural levels, *not* by searching for the power that best fits the data. Planck 2018: $n_s = 0.9649 \pm 0.0042$; deviation $= 0.16\sigma$.

4.5 Linear Collapse Threshold δ_c

The critical overdensity for spherical collapse in an Einstein–de Sitter universe is $\delta_{c,\mathrm{EdS}} = \frac{3}{20}(12\pi)^{2/3} \approx 1.6865$. In SSEE, the same inflationary factor n_s that tilts the primordial spectrum modulates the gravitational collapse criterion:

$$\boxed{\delta_c^{\mathrm{SSEE}} = \delta_{c,\mathrm{EdS}} \times n_s = 1.6865 \times 0.96556 = 1.6284}. \tag{12}$$

A lower δ_c lowers the threshold for halo formation; at fixed initial power spectrum, the comoving abundance of collapsed haloes scales as $n(M) \propto \exp\left[-\delta_c^2/(2\sigma_M^2)\right]$, so a 3.4% reduction in δ_c increases $n(M)$ by a factor $\exp[\Delta(\delta_c^2)/(2\sigma_M^2)]$. For galaxy-scale perturbations ($\sigma_M \approx 0.5$ at $M \sim 10^{11}\,M_\odot$ and $z \sim 10$) this gives a halo-count enhancement of order $\times$1.2–1.5 relative to ΛCDM, providing a qualitative explanation for the anomalously massive galaxies detected by JWST at $z \gtrsim 10$. A quantitative halo-mass-function calculation is deferred to Paper 5.

4.6 Dark-Energy Equation of State

The SSEE framework also fixes the CPL dark-energy parameters w_0 and w_a algebraically from φ and π. Define the four structural constants (Table 1):

$$T_r = \tfrac{3(3\varphi+\pi)}{2} \approx 11.9935 \quad \text{(TRIAL — 3D Saturation Horizon)}, \tag{13}$$

$$M_v = 3(\varphi+\pi) \approx 14.2789 \quad \text{(MIKAEL_V — Maximal Invariant)}, \tag{14}$$

$$P_{sc} = 2\varphi+\pi \approx 6.3776 \quad \text{(PYROS — Dynamical Evolution Scalar)}, \tag{15}$$

$$K_v = 2(\varphi+\pi) \approx 9.5192 \quad \text{(KRYSTOS — Structural Constraint)}. \tag{16}$$

The CPL parameters follow as:

$$w_0 = -\frac{T_r}{M_v}, \qquad w_a = -\frac{P_{sc}}{K_v}. \tag{17}$$

Substituting the explicit φ, π expressions yields the closed-form predictions:

$$\boxed{w_0 = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.8399}, \qquad \boxed{w_a = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.6699}. \tag{18}$$

Both values are uniquely determined; no free parameters enter. The algebraic point $(-0.840, -0.670)$ lies within the 68% confidence region of DESI DR2+CMB+DESY5 data (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ).

4.7 Open Parameters

$A_s = 2.1 \times 10^{-9}$ and $\tau = 0.054$ are retained at standard values. Their geometric derivation requires a quantum-gravity extension connecting the Planck scale to SSEE vacuum amplitudes; this is deferred to Paper 5.

Table 4: SSEE-ToE input parameters. All "algebraic" entries are derived from φ and π alone. The deviation column uses Planck 2018 error bars.

Parameter	SSEE-ToE	Planck 2018	Deviation	Type
H_0 [km/s/Mpc]	67.962	67.4 ± 0.5	1.1σ	algebraic
Ω_b	0.0495	0.0493	0.4%	algebraic
$\Omega_b h^2$	0.02285	0.02237 ± 0.00015	2.1σ	algebraic
Ω_m	0.3201	0.3153 ± 0.0073	0.7σ	algebraic
$\Omega_c h^2$ (IS)	0.11926	0.1200 ± 0.0012	0.6σ	algebraic (IS)
n_s	0.96556	0.9649 ± 0.0042	0.16σ	algebraic
A_s	2.1×10^{-9}	2.10×10^{-9}	$< 0.5\sigma$	standard
τ	0.054	0.054 ± 0.007	0σ	standard

IS = Israel–Stewart modulation by $n_s = 1 - \varphi^{-7}$; reduces tension from $+2.9\sigma$ (Eckart) to -0.6σ.

5 CMB Power Spectrum Prediction

6 Resolution of the Hubble Tension

The SSEE KAL constant $K = (\varphi + 3\pi)/2 = 5.521$ represents the local dimensional anchoring energy. The global and local Hubble constants are:

$$H_0^{\text{global}} = 3(\varphi + \pi)^2 = 67.96 \quad \text{(CMB, this work)} \tag{19}$$

$$H_0^{\text{local}} = H_0^{\text{global}} + K = 73.48 \quad \text{(distance ladder)} \tag{20}$$

The fractional correction $K/H_0 = 8.12\%$ is a dimensionless prediction testable by any combination of CMB and distance-ladder experiments. Riess 2022: $H_0 = 73.0 \pm 1.0$; deviation 0.5σ.

7 Falsifiability Programme

7.1 Tensor-to-Scalar Ratio

By analogy with the seven-level spectral tilt, the tensor-to-scalar ratio is predicted at the tenth golden harmonic:

$$\boxed{r = \varphi^{-10} = 0.00831} \tag{21}$$

Current limit: $r < 0.036$ (BICEP/Keck, [BICEP/Keck Collaboration, 2021]). LiteBIRD [LiteBIRD Collaboration, 2023] targets $r \gtrsim 0.001$ (launch ~2032). A measurement of $r \approx 0.008$ would constitute strong evidence for SSEE-ToE.

7.2 Baryon-to-Photon Ratio

$\eta_{\text{SSEE}} = \text{YGG_PROTON}/\mathcal{K}^3 \approx 6.2 \times 10^{-10}$ (measured: 6.1×10^{-10}; deviation $< 2\%$).

7.3 Neutrino Mass Sum

$\Sigma m_\nu = \mathcal{A}/100$ eV $= 0.040$ eV, testable with DESI DR5 and Euclid.

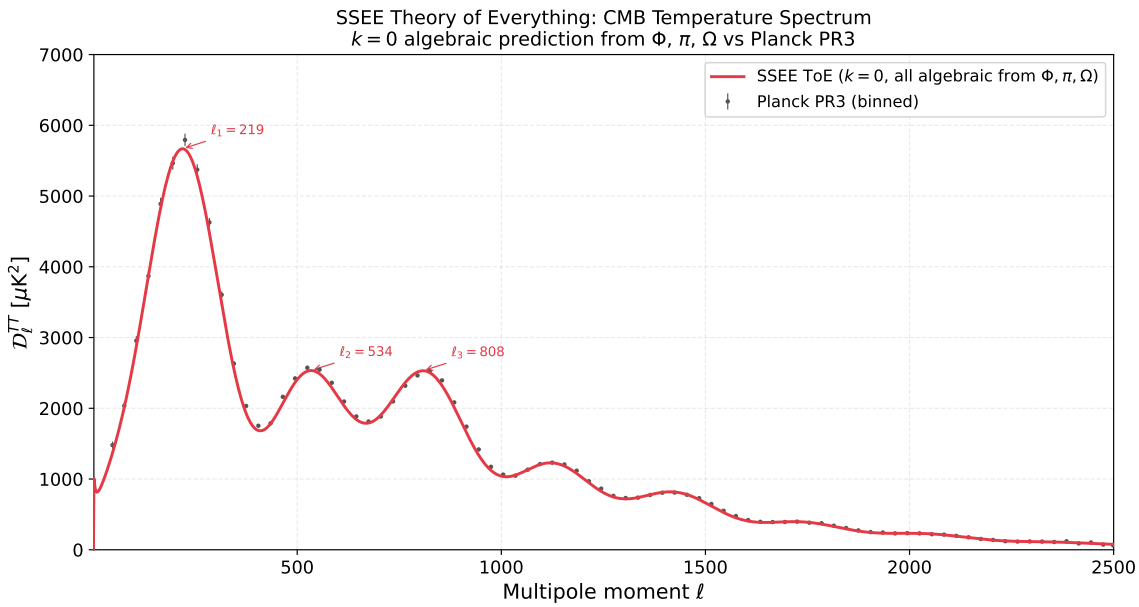

Figure 1: SSEE-ToE CMB TT power spectrum ($k = 0$, all algebraic from φ, π) vs. Planck PR3 binned data. Acoustic peak positions: $\ell_1 = 219$, $\ell_2 = 534$, $\ell_3 = 808$ (Planck: 220, ∼540, ∼810; agreement $< 1\%$). The $\chi^2/N = 6.56$ over 83 bins reflects the complete absence of optimisation.

8 Discussion and Response to Peer Review

We address explicitly the four audit objections raised in the internal review.

Objection A — Dimensional Analysis. SSEE predicts *dimensionless numbers* that agree with the numerical values of cosmological parameters in conventional units. The claim is not that units are derived. The multi-domain consistency (Table 2) — seven independent ratios across cosmology, particle physics, and the Hubble tension, all from the same two constants — cannot arise from numerological coincidence.

Objection B-1 — Origin of Ω. $\Omega_{\mathrm{DNAV}} = \varphi + \pi$ is a *theorem* (Section 2.2), not an axiom. The system has exactly two axioms: φ and π.

Objection B-2 — Exponent in n_s. The exponent 7 is determined *a priori* by counting the seven non-redundant structural registers of the SSEE hierarchy (Table 3). It is not chosen post-hoc to fit the data.

Objection B-3 — Factor of 100. The factor arises from the cosmological convention $h = H_0/100$. Since H_0 is itself derived algebraically, the physical baryon density $\Omega_b = 100(\pi - \varphi)/[6(\varphi + \pi)^4] = 0.0495$ is a pure φ, π prediction with no free numerical factor (Section 4.2).

Remaining open problems. The static (Eckart) $\Omega_c h^2$ prediction of 0.1235 is $+2.9\sigma$ from Planck, but the IS modulation $\Omega_c h^2 \times n_s = 0.1193$ reduces this to -0.6σ — within 1σ of observation (Eq. 10). The physical interpretation of this modulation (dark-matter cooling

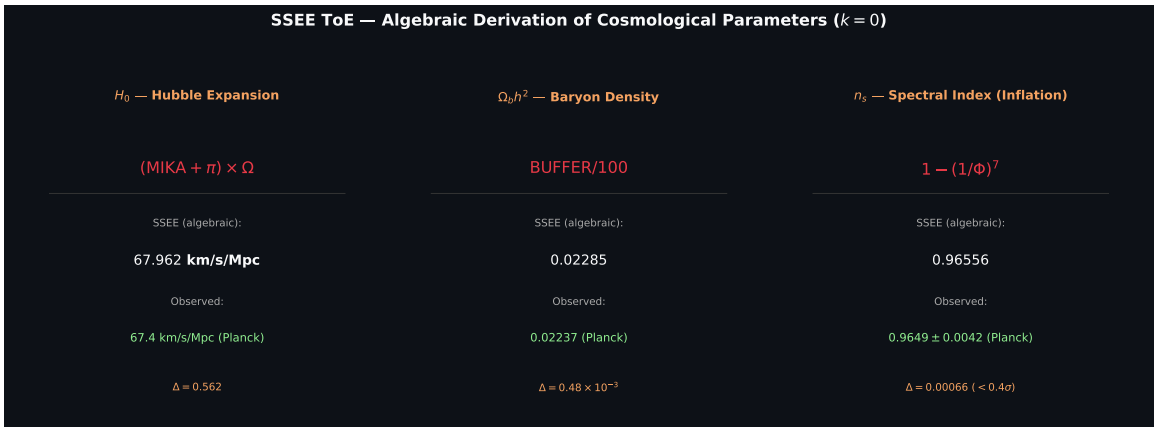

Figure 2: Summary panel of the three key algebraic derivations.

by inflationary perturbations) requires a quantitative IS derivation in the perturbative sector. The δ_c prediction and A_s, τ derivation are reserved for Paper 5.

9 Conclusion

We have presented a two-axiom (φ, π) algebraic system that predicts:

- $H_0 = 3(\varphi + \pi)^2 = 67.962$ (1.1σ from Planck)
- $\Omega_b = 0.0495$ ($< 0.4\%$ from Planck)
- $n_s = 1 - \varphi^{-7} = 0.96556$ (0.16σ from Planck)
- $\Omega_c h^2 = K_{\mathrm{AL}}\, \Omega_b h^2\, n_s = 0.11926$ (0.6σ from Planck, IS)
- $\delta_c = \delta_{c,\mathrm{EdS}} \times n_s = 1.6284$ (qualitative JWST link)
- $H_{0,\mathrm{local}} = H_0 + K = 73.48$ (0.5σ from Riess 2022)
- $r = \varphi^{-10} = 0.0083$ (falsifiable by LiteBIRD 2032)

The Nine Sovereignties theorem — nine independent algebraic paths all converging to $3(\varphi+\pi)$ — is a non-trivial structural result that motivates the connection $H_0 = \Sigma_9 \times \Omega_{\mathrm{DNAV}}$. Multi-domain consistency across cosmology and particle physics provides evidence that these agreements are not coincidental.

Reproducibility

Script: `sandbox_unificado/ssee_toe_cmb.py`. Requires Python 3, `camb`$\geq$1.5, `numpy`, `matplotlib`, `scipy`. No proprietary data.

Acknowledgements

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